

Faculdade de Ciências - Departamento de Computação

Disciplina: Pesquisa Operacional

Docente: Douglas Rodrigues

Lista de Exercícios - L5

Método Branch & Bound

Instruções:

- A lista é individual.
- A resolução deve ser feita de próprio punho (manuscrita).
- A lista deve ser entregue no formato PDF.
- O arquivo deve ser renomeado com o nome e sobrenome seguidos pela letra “L” e o número da lista.
- Deve-se utilizar letras minúsculas separados por traço baixo (underline).
- **Exemplo:** douglas_rodrigues_l5.pdf

Exercícios

1. Utilize o método de Bifurcação e Limite (Branch and Bound) para resolver os seguintes problemas:

(a)

$$\begin{aligned} \max z &= 1x_1 + 9x_2 + x_3 \\ \text{sujeito a: } &\begin{cases} x_1 + 2x_2 + 3x_3 \leq 9 \\ 3x_1 + 2x_2 + 2x_3 \leq 15 \\ \vec{x} \in \mathbb{Z}_+^3 \end{cases} \end{aligned}$$

(b)

$$\begin{aligned} \max z &= 7x_1 + 7x_2 + 6x_3 + 9x_4 \\ \text{sujeito a: } &\begin{cases} 4x_1 + 5x_2 + 3x_3 + 5x_4 \leq 300 \\ 2x_1 + 15x_2 + 3x_3 + 3x_4 \leq 200 \\ \vec{x} \in \mathbb{Z}_+^4 \end{cases} \end{aligned}$$

(c)

$$\begin{array}{l} \max z = 1x_1 + 2x_2 + 3x_3 + 1x_4 \\ \text{sujeito a: } \begin{cases} 3x_1 + 2x_2 + 1x_3 + 4x_4 \leq 10 \\ 5x_1 + 3x_2 + 2x_3 + 5x_4 \leq 5 \\ \vec{x} \in \mathbb{Z}_+^4 \end{cases} \end{array}$$

Lista de Exercícios 5 - Método Branch & Bound

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$$\alpha) \max z = 1x_1 + 9x_2 + x_3$$

$$\text{s.a.: } \begin{cases} x_1 + 2x_2 + 3x_3 \leq 9 \\ 3x_1 + 2x_2 + 2x_3 \leq 15 \\ \vec{x} \geq \mathbb{Z}_+^3 \end{cases}$$

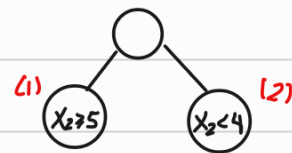
$$\min z' = -x_1 - 9x_2 - x_3$$

$$\text{s.a.: } \begin{cases} x_1 + 2x_2 + 3x_3 + x_4 = 9 \\ 3x_1 + 2x_2 + 2x_3 + x_5 = 15 \\ \vec{x} \geq \mathbb{Z}_+^3 \end{cases}$$

	x_1	x_2	x_3	x_4	x_5	b	
$\rightarrow x_4$	1	2	3	1	0	9	$L_1 = L_1 \div 2$
x_5	3	2	2	0	1	15	$L_2 = L_2 - 2L_1$
	-1	-9	-1	0	0	0	$L_3 = L_2 + 9L_1$

$\min (9/2, 15/2)$

	x_1	x_2	x_3	x_4	x_5	b	
x_2	$1/2$	1	$3/2$	$1/2$	0	$9/2$	
x_5	2	0	-1	-1	1	6	
	$7/2$	0	$25/2$	$9/2$	0	$81/2$	

$$\begin{matrix} -1 & -9 & -1 & 0 & 0 & 0 \\ 9/2 & 9 & 29/2 & 9/2 & 0 & 81/2 \end{matrix}$$


nó 1: $x_2 \geq 5$

testando as outras restrições c/ $x_2 = 5$

$$\text{s.a.: } \begin{cases} x_1 + 2x_2 + 3x_3 \leq 9 \rightarrow x_1 + 10 + x_3 \leq 9 \text{ (impossível)} \\ 3x_1 + 2x_2 + 2x_3 \leq 15 \\ \vec{x} \geq \mathbb{Z}_+^3 \end{cases}$$

$\therefore$ problema inviável

Nó 2: ($x_2 < 4$)

testando as outras restrições c/ $x_2 = 4$

$$\begin{aligned} \max z &= 1x_1 + 9x_2 + x_3 \rightarrow \max x_1 + x_3 + 36 \rightarrow \min -x_1 - x_3 \\ \text{s.a.: } \begin{cases} x_1 + 2x_2 + 3x_3 \leq 9 \rightarrow x_1 + 8 + 3x_3 \leq 9 \rightarrow x_1 + 3x_3 \leq 1 \rightarrow \\ 3x_1 + 2x_2 + 2x_3 \leq 15 \rightarrow 3x_1 + 8 + 2x_3 \leq 15 \rightarrow 3x_1 + 2x_3 \leq 7 \rightarrow \\ \vec{x} \geq \mathbb{Z}_+^3 \end{cases} & \begin{cases} x_1 + 3x_3 + x_4 = 1 \\ 3x_1 + 2x_3 + x_5 = 7 \end{cases} \end{aligned}$$

	x_1	x_3	x_4	x_5	b	
$\rightarrow x_4$	1	3	1	0	1	$L_1 = L_1$
x_5	3	2	0	1	7	$L_2 = L_2 - 3L_1$
	-1	-1	0	0	36	$L_3 = L_3 + L_1$

$\min(1, \frac{3}{2})$

	x_1	x_3	x_4	x_5	b
x_1	1	1	1	0	1
x_5	0	-1	-3	1	4
	0	0	1	0	37

coef. VNB $\geq 0 \therefore$ solução ótima

$$x^* = (1 \ 4 \ 0 \ 0 \ 4) \ 0 \quad z^* = 37 //$$

b) $\max z = 7x_1 + 7x_2 + 6x_3 + 9x_4$ $\min z = -7x_1 - 7x_2 - 6x_3 - 9x_4$

$$\text{s.a.: } \begin{cases} 4x_1 + 5x_2 + 3x_3 + 5x_4 \leq 300 \rightarrow \\ 2x_1 + 15x_2 + 3x_3 + 3x_4 \leq 200 \\ \vec{x} \in \mathbb{Z}_+^4 \end{cases} \quad \text{s.a.: } \begin{cases} 4x_1 + 5x_2 + 3x_3 + 5x_4 + x_5 = 300 \\ 2x_1 + 15x_2 + 3x_3 + 3x_4 + x_6 = 200 \\ \vec{x} \in \mathbb{Z}_+^4 \end{cases}$$

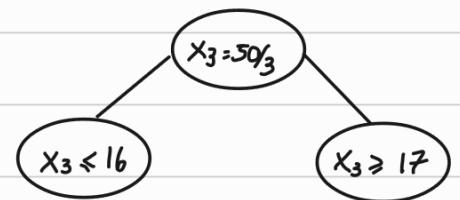
	x_1	x_2	x_3	x_4	x_5	x_6	b	
x_5	4	5	3	5	1	0	300	$L_1 = L_1 \div 5$
x_6	2	15	3	3	0	1	200	$L_2 = L_2 - 3L_1$
Z'	-7	-7	-6	-9	0	0	0	$L_3 = L_3 + 9L_1$

$\min (60, 200/3)$

	x_1	x_2	x_3	x_4	x_5	x_6	b	
x_4	$4/5$	1	$3/5$	1	$1/5$	0	60	$L_1 = L_1 - 3/5 L_2$
x_6	$-2/5$	12	$6/5$	0	$-3/5$	1	20	$L_2 = L_2 \cdot 5/6$
Z'	$1/5$	2	$-3/5$	0	$9/5$	0	540	$L_3 = L_3 + 3/5 L_2$

$\min (100, 50/3)$

	x_1	x_2	x_3	x_4	x_5	x_6	b
x_4	1	-5	0	1	$1/2$	$-1/2$	50
x_3	$-1/3$	10	1	0	$-1/2$	$5/6$	$50/3$
Z'	0	8	0	0	$3/2$	$1/2$	550



$$x_3 \leq 16: \quad x_3 \leq 16$$

$$x_3 = 50/3 - (-1/3 x_1 + 10x_2 - 1/2 x_5 + 5/6 x_6) \leq 16$$

$$x_3 = 1/3 x_1 - 10x_2 + 1/2 x_5 - 5/6 x_6 \leq -2/3$$

$$-1/3 x_1 + 10x_2 - 1/2 x_5 + 5/6 x_6 \geq 2/3 \rightarrow -1/3 x_1 + 10x_2 - 1/2 x_5 + 5/6 x_6 - x_7 + w_1 = 2/3$$

$$\min Z^a = w_1 = 1/3 x_1 - 10x_2 + 1/2 x_5 - 5/6 x_6 + x_7$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_1	b
x_4	1	-5	0	1	$1/2$	$-1/2$	0	0	50 $L_1 = L_1 + 5L_3$
x_3	$-1/3$	10	1	0	$-1/2$	$5/6$	0	0	$50/3$ $L_2 = L_2 - 10L_3$
w_1	$-1/3$	10	0	0	$-1/2$	$5/6$	-1	1	$2/3$ $L_3 = L_3 \div 10$
Z^a	$1/3$	-10	0	0	$1/2$	$-5/6$	1	0	$Z^a - 2/3$ $L_4 = L_4 + 10L_3$
Z'	0	8	0	0	$3/2$	$1/2$	0	0	550 $L_5 = L_5 - 8L_3$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
x_4	$5/6$	0	0	1	$1/4$	$-1/2$	$-1/2$	$15\frac{1}{3}$
x_3	0	0	1	0	0	0	1	16
x_2	$-1/30$	1	0	0	$-1/20$	$1/12$	$-1/10$	$1/15$
z'	$4/15$	0	0	0	$19/10$	$-1/6$	$4/5$	$8242/15$

nó 2: $x_3 \geq 17$

$$x_3 = \frac{1}{3}x_1 - 10x_2 + \frac{1}{2}x_5 - \frac{5}{6}x_6 \geq \frac{1}{3}$$

$$\frac{1}{3}x_1 - 10x_2 + \frac{1}{2}x_5 - \frac{5}{6}x_6 - x_7 + w_1 = \frac{1}{3}$$

$$\min z'' = w_1 = \frac{1}{3} - \frac{1}{3}x_1 + 10x_2 - \frac{1}{2}x_5 + \frac{5}{6}x_6 + x_7$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_1	b	
x_4	1	-5	0	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	50	$L_1 = L_1 - \frac{1}{2}L_3'$
x_3	$-\frac{1}{3}$	10	1	0	$-\frac{1}{2}$	$\frac{5}{6}$	0	0	$\frac{50}{3}$	$L_2 = L_1 + \frac{1}{2}L_3'$
$\rightarrow w_1$	$\frac{1}{3}$	-10	0	0	$\frac{1}{2}$	$-\frac{5}{6}$	-1	1	$\frac{1}{3}$	$L_3 = L_3 \cdot 2$
z''	$-\frac{1}{3}$	10	0	0	$-\frac{1}{2}$	$\frac{5}{6}$	1	0	$z'' - \frac{1}{3}$	$L_4 = L_4 + \frac{1}{2}L_3'$
z'	0	8	0	0	$\frac{3}{2}$	$\frac{1}{2}$	0	0	550	$L_5 = L_5 - \frac{3}{2}L_3'$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
x_4	$2/3$	5	0	1	0	$1/3$	1	$149/3$
x_3	0	0	1	0	0	0	-1	17
$\rightarrow x_5$	$2/3$	-20	0	0	1	$-5/3$	-2	$2/3$
z'	-1	38	0	0	0	3	3	549

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
x_4	0	25	0	1	-1	$17/6$	3	49
x_3	0	0	1	0	0	0	-1	17
x_1	1	-30	0	0	$3/2$	$-5/2$	-3	1
z'	0	8	0	0	$3/2$	$4/3$	0	550

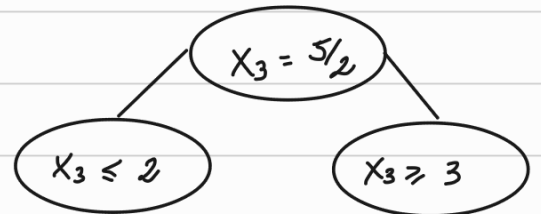
como no nó 1 $z^* = 8242/15$ e possui variáveis fracionárias, esse nó será podado por limite, já que esse z^* já é menor do que o do nó 2.

$$x^* = (1 \ 0 \ 17 \ 49 \ 0 \ 0 \ 0) \text{ e } z^* = 560 //$$

c) $\max \quad z = x_1 + 2x_2 + 3x_3 + x_4$
s.a.: $\begin{cases} 3x_1 + 2x_2 + x_3 + 4x_4 \leq 10 \\ 5x_1 + 3x_2 + 2x_3 + 5x_4 \leq 5 \\ \vec{x} \in \mathbb{Z}_+^4 \end{cases} \rightarrow \begin{cases} \min z' = -x_1 - 2x_2 - 3x_3 - x_4 \\ 3x_1 + 2x_2 + x_3 + 4x_4 + x_5 = 10 \\ 5x_1 + 3x_2 + 2x_3 + 5x_4 + x_6 = 5 \\ \vec{x} \in \mathbb{Z}_+^6 \end{cases}$

	x_1	x_2	x_3	x_4	x_5	x_6	b	
x_5	3	2	1	4	1	0	10	$\min(10, 2)$
x_6	5	3	2	5	0	1	5	$L_1 = L_1 - L_2'$
z	-1	-2	-3	-1	0	0	0	$L_2 = L_2 \div 2$
								$L_3 = L_3 + 3L_2'$

	x_1	x_2	x_3	x_4	x_5	x_6	b
x_5	$1/2$	$1/2$	0	$3/2$	1	$-1/2$	$15/2$
x_3	$5/2$	$3/2$	1	$5/2$	0	$1/2$	$5/2$
z	$13/2$	$5/2$	0	$13/2$	0	$3/2$	$15/2$



$$x_3 \leq 2:$$

$$5/2 - 5/2x_1 - 3/2x_2 - 5/2x_4 - 1/2x_6 \leq 2$$

$$5/2x_1 + 3/2x_2 + 5/2x_4 + 1/2x_6 \geq 1/2$$

$$5x_1 + 3x_2 + 5x_4 + x_6 \geq 1$$

$$5x_1 + 3x_2 + 5x_4 + x_6 - x_7 + w_1 = 1$$

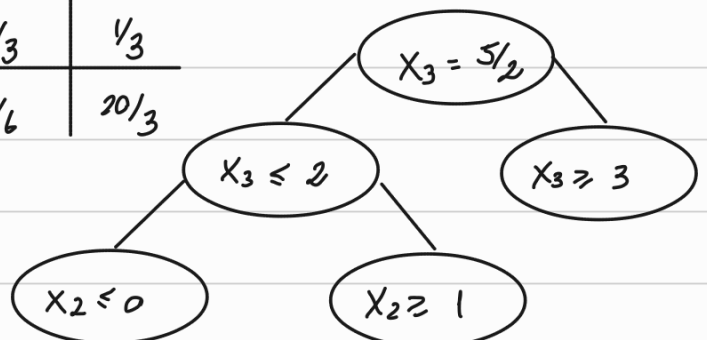
$$\min z^a = w_1 = 1 - 5x_1 - 3x_2 - 5x_4 - x_6 + x_7$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_1	b	$\min(15, 1, \frac{1}{5})$
x_5	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{3}{2}$	1	$-\frac{1}{2}$	0	0	$\frac{15}{2}$	$L_1 = L_1 - \frac{1}{2}L_3'$
x_3	$\frac{5}{2}$	$\frac{3}{2}$	1	$\frac{5}{2}$	0	$\frac{1}{2}$	0	0	$\frac{5}{2}$	$L_2 = L_2 - \frac{5}{2}L_3'$
w_1	5	3	0	5	0	1	-1	1	1	$L_3 = L_3 \div 5$
Z^a	-5	-3	0	-5	0	-1	1	0	$Z^a - 1$	$L_4 = L_4 + 5L_3'$
Z	$\frac{13}{2}$	$\frac{5}{2}$	0	$\frac{13}{2}$	0	$\frac{3}{2}$	0	0	$\frac{15}{2}$	$L_5 = L_5 - \frac{13}{2}L_3'$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_1	b
x_5	0	$\frac{1}{5}$	0	1	1	$-\frac{3}{5}$	$\frac{1}{10}$	$-\frac{1}{10}$	$\frac{37}{5}$
x_3	0	0	1	0	0	0	$\frac{1}{2}$	$-\frac{1}{2}$	2
x_1	1	$\frac{3}{5}$	0	1	0	$\frac{1}{5}$	$-\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$
Z^a	0	0	0	0	0	0	0	1	Z^a
Z	0	$-\frac{7}{5}$	0	0	0	$\frac{1}{5}$	$\frac{13}{10}$	$-\frac{13}{10}$	$\frac{31}{5}$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b	$\min(37, 3)$
x_5	0	$\frac{1}{5}$	0	1	1	$-\frac{3}{5}$	$\frac{1}{10}$	$\frac{37}{5}$	$L_1 = L_1 - \frac{1}{5}L_3'$
x_3	0	0	1	0	0	0	$\frac{1}{2}$	2	$L_2 = L_2$
x_1	1	$\frac{3}{5}$	0	1	0	$\frac{1}{5}$	$-\frac{1}{5}$	$\frac{1}{5}$	$L_3 = L_3 \cdot \frac{5}{3}$
Z	0	$-\frac{7}{5}$	0	0	0	$\frac{1}{5}$	$\frac{13}{10}$	$\frac{31}{5}$	$L_4 = L_4 + \frac{7}{5}L_3'$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
x_5	$-\frac{1}{3}$	0	0	$\frac{2}{3}$	1	$-\frac{2}{3}$	$\frac{1}{6}$	$\frac{22}{3}$
x_3	0	0	1	0	0	0	$\frac{1}{2}$	2
x_2	$\frac{5}{3}$	1	0	$\frac{5}{3}$	0	$\frac{1}{3}$	$-\frac{1}{3}$	$\frac{1}{3}$
Z	$\frac{7}{3}$	0	0	$\frac{2}{3}$	0	$\frac{2}{3}$	$\frac{5}{6}$	$\frac{20}{3}$



$$x_3 \geq 3:$$

$$5/2 - 5/2 x_1 - 3/2 x_2 - 5/2 x_4 - 1/2 x_6 \geq 3$$

$$- 5/2 x_1 - 3/2 x_2 - 5/2 x_4 - 1/2 x_6 \geq 1/2$$

$$- 5/2 x_1 - 3/2 x_2 - 5/2 x_4 - 1/2 x_6 - x_7 + w_1 = 1/2$$

$$- 5x_1 - 3x_2 - 5x_4 - x_6 - x_7 + w_1 = 1$$

$$\min z^a = w_1 = 1 + 5x_1 + 3x_2 + 5x_4 + x_6 + x_7$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_1	b
x_5	$1/2$	$1/2$	0	$3/2$	1	$-1/2$	0	0	$15/2$
x_3	$5/2$	$3/2$	1	$5/2$	0	$1/2$	0	0	$5/2$
w_1	-5	-3	0	-5	0	-1	1	0	1
z^a	5	3	0	5	0	1	1	0	$z^a - 1$
z	$13/2$	$5/2$	0	$13/2$	0	$3/2$	0	0	$15/2$

A coef VNB $\leq 0 \therefore$ não é possível zerar z^a
problema infactível

$$x_2 \geq 1: \text{ ramo inviável}$$

$$x_2 = 0: 1/3 - 5/3 x_1 - 5/3 x_4 - 1/3 x_6 + 1/3 x_7 = 0$$

$$5/3 x_1 + 5/3 x_4 + 1/3 x_6 - 1/3 x_7 = 1/3$$

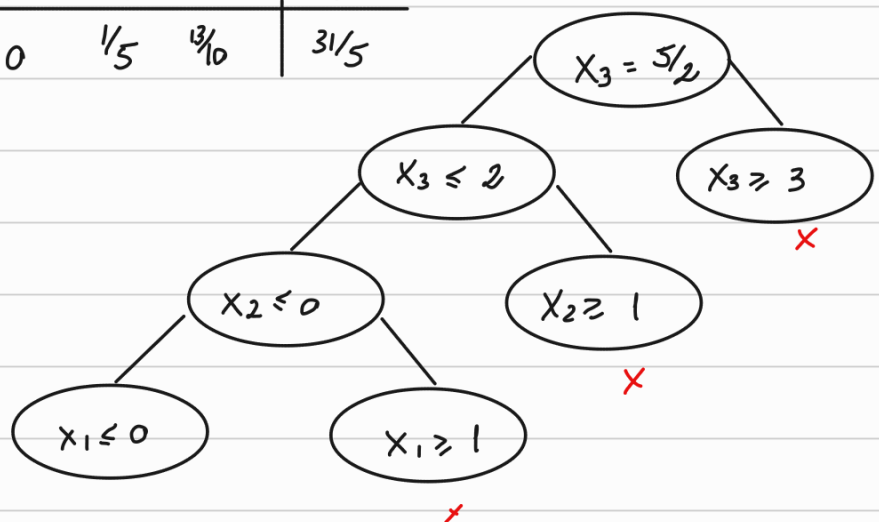
$$5/3 x_1 + 5/3 x_4 + 1/3 x_6 - 1/3 x_7 + w_2 = 1/3$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_2	b
x_5	$-1/3$	0	0	$2/3$	1	$-2/3$	$1/6$	0	$22/3$
x_3	0	0	1	0	0	0	$1/2$	0	2
x_2	$5/3$	1	0	$5/3$	0	$1/3$	$-1/3$	0	$1/3$
w_2	$5/3$	0	0	$5/3$	0	$1/3$	$-1/3$	1	$1/3$
z^a	$-5/3$	0	0	$-5/3$	0	$-1/3$	$1/3$	0	$z^a - 1/3$
z	$7/3$	0	0	$7/3$	0	$2/3$	$5/6$	0	$20/3$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_2	b	
x_5	$-\frac{1}{3}$	0	0	$\frac{2}{3}$	1	$-\frac{2}{3}$	$\frac{1}{6}$	0	$\frac{22}{3}$	$L_1 = L_1 + \frac{1}{3}L_4'$
x_3	0	0	1	0	0	0	$\frac{1}{2}$	0	2	$L_2 = L_2$
x_2	$\frac{5}{3}$	1	0	$\frac{5}{3}$	0	$\frac{1}{3}$	$-\frac{1}{3}$	0	$\frac{1}{3}$	$L_3 = L_3 - \frac{5}{3}L_4'$
w_2	$\frac{5}{3}$	0	0	$\frac{5}{3}$	0	$\frac{1}{3}$	$-\frac{1}{3}$	1	$\frac{1}{3}$	$L_4 = L_4 \cdot \frac{3}{5}$
z^a	$-\frac{5}{3}$	0	0	$-\frac{5}{3}$	0	$-\frac{1}{3}$	$\frac{1}{3}$	0	$z^a - \frac{1}{3}$	$L_5 = L_5 + \frac{5}{3}L_4'$
z	$\frac{7}{3}$	0	0	$\frac{7}{3}$	0	$\frac{2}{3}$	$\frac{5}{6}$	0	$\frac{20}{3}$	$L_6 = L_6 - \frac{7}{3}L_4'$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_2	b	
x_5	0	0	0	1	1	$-\frac{3}{5}$	$\frac{1}{10}$	1	$65\frac{3}{75}$	
x_3	0	0	1	0	0	0	$\frac{1}{2}$	0	2	
x_2	0	1	0	0	0	0	0	$-\frac{5}{3}$	0	
x_1	1	0	0	1	0	$\frac{1}{5}$	$-\frac{1}{5}$	1	$\frac{1}{5}$	
z^a	0	0	0	0	0	0	0	1	z^a	
z	0	0	0	0	0	$\frac{1}{5}$	$\frac{13}{10}$	$-\frac{7}{3}$	$31\frac{1}{5}$	

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b	
x_5	0	0	0	1	1	$-\frac{3}{5}$	$\frac{1}{10}$	$65\frac{3}{75}$	
x_3	0	0	1	0	0	0	$\frac{1}{2}$	2	
x_2	0	1	0	0	0	0	0	0	
x_1	1	0	0	1	0	$\frac{1}{5}$	$-\frac{1}{5}$	$\frac{1}{5}$	
z	0	0	0	0	0	$\frac{1}{5}$	$\frac{13}{10}$	$31\frac{1}{5}$	



$$x_1 \geq 1:$$

$$x_3 \leq 2 \quad 3(1) + 2(0) + x_3 + 4x_4 \leq 10$$

$$x_2 \leq 0 \quad 5(1) + 3(0) + 2(0) + 5x_4 \leq 5 \quad \times$$

$$x_1 \geq 1$$

ramo inviável

$$x_1 \leq 0:$$

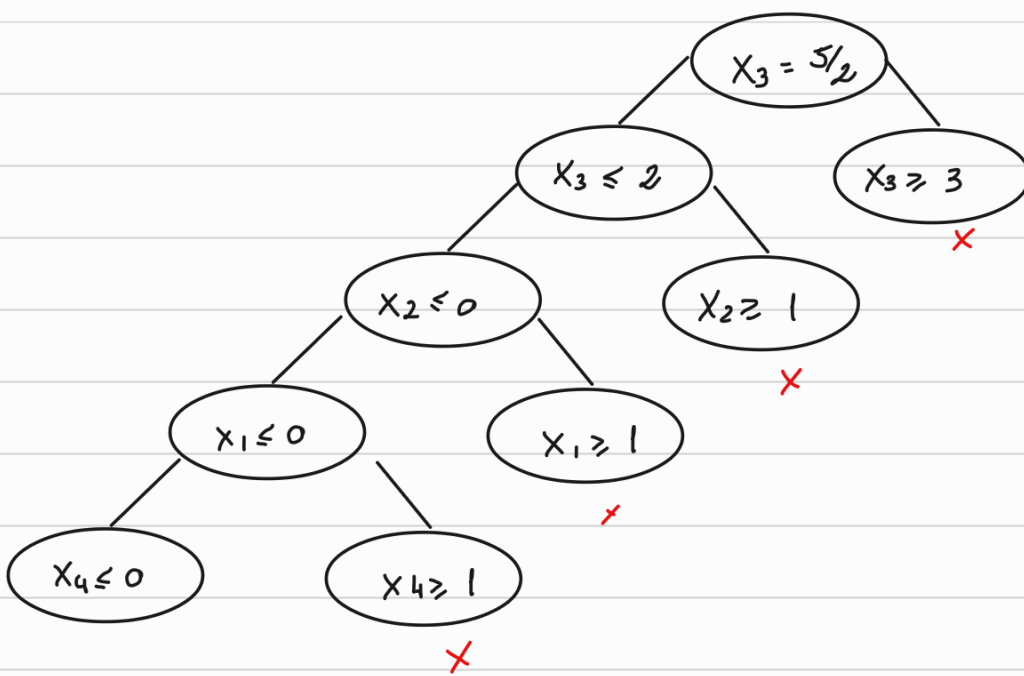
$$x_4 + \frac{1}{5}x_6 - \frac{1}{5}x_7 = \frac{1}{5}$$

$$x_4 + \frac{1}{5}x_6 - \frac{1}{5}x_7 + w_3 = \frac{1}{5}$$

$$\text{Min } z^a = w_3 = \frac{1}{5} - x_4 - \frac{1}{5}x_6 + \frac{1}{5}x_7$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_3	b
x_5	0	0	0	1	1	$-\frac{3}{5}$	$\frac{1}{10}$	0	$6\frac{3}{25} \quad L_1 = L_1 - L_5$
x_3	0	0	1	0	0	0	$\frac{1}{2}$	0	2 $L_2 = L_2$
x_2	0	1	0	0	0	0	0	0	0 $L_3 = L_3$
x_1	1	0	0	1	0	$\frac{1}{5}$	$-\frac{1}{5}$	0	$\frac{1}{5} \quad L_4 = L_4 - L_5$
$\rightarrow w_3$	0	0	0	1	0	$\frac{1}{5}$	$-\frac{1}{5}$	1	$\frac{1}{5} \quad L_5 = L_5$
z^a	0	0	0	-1	0	$-\frac{1}{5}$	$\frac{1}{5}$	0	$z^a - \frac{1}{5} \quad L_6 = L_6 + L_5$
z	0	0	0	0	0	$\frac{1}{5}$	$\frac{3}{10}$	0	$\frac{31}{5} \quad L_7 =$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_3	b
x_5	0	0	0	0	1	$-\frac{6}{5}$	$-\frac{3}{5}$	1	$\frac{36}{5}$
x_3	0	0	1	0	0	0	$\frac{1}{2}$	0	2
x_2	0	1	0	0	0	0	0	0	0
x_1	1	0	0	0	0	0	0	0	0
x_4	0	0	0	1	0	$\frac{1}{5}$	$-\frac{1}{5}$	1	$\frac{1}{5}$
z^a	0	0	0	0	0	0	0	1	z^a
z	0	0	0	0	0	$\frac{1}{2}$	$\frac{11}{5}$	0	$\frac{31}{5}$



$$x^* = (0 \ 0 \ 2 \ 0) \text{ e } z^* = 6 //$$